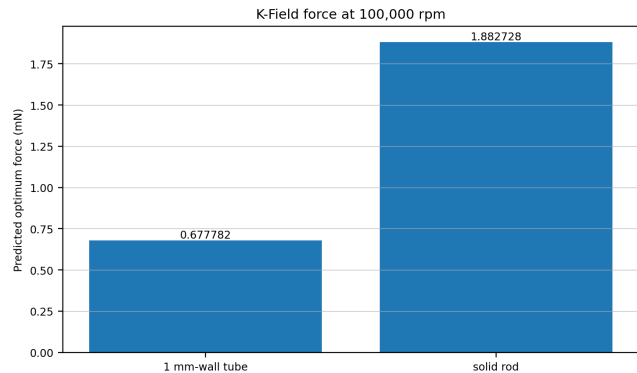
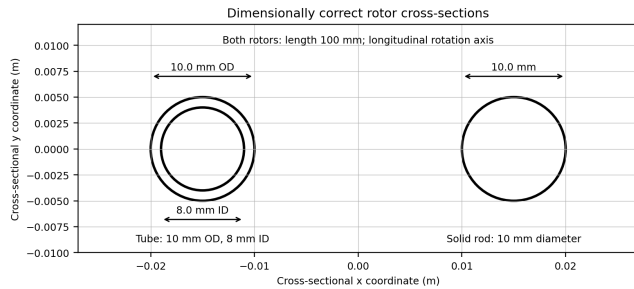


Solid Rod Versus Tube

Mass-linear force, specific-force invariance, polar inertia, stored energy, and hollowing laws for equal-diameter carbon-fiber rotors



Title figures. The cross-sections use the exact 10 mm outside diameter, 8 mm tube inside diameter, and 100 mm length. The force bars are generated from the complete numerical ledger at 100,000 rpm and the optimum Earth-CMB-relative orientation.

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This paper is a standalone educational and archival reference. It defines the Cell3 geometry and field inputs used by the rotating-body force law and compares two carbon-fiber rotors having the same 10 mm outside diameter and 100 mm length: a tube with a 1 mm radial wall and a solid rod. Every numerical table and figure is generated from the machine-readable companion ledger.

Abstract

Under the current integrated rotating-body law, the optimum force of a complete axisymmetric rotor is proportional to rotating mass and signed angular speed. The force per unit rotating mass is therefore independent of whether the circular cross-section is hollow or solid when both bodies rotate at the same angular speed and have the same material coupling. Using a working carbon-fiber density of 1600 kg/m^3 , the 10 mm outside-diameter, 8 mm inside-diameter tube has a mass of 4.523893 g and generates 0.677782 mN at 100,000 rpm. The solid rod has a mass of 12.566371 g and generates 1.882728 mN. The solid rod produces 2.777778 times as much force because it contains 2.777778 times as much rotating mass. Both bodies generate the same specific force, 0.149822768 N/kg.

The polar moment and stored spin energy do not scale in the same proportion as mass. The tube retains 59.04% of the solid rod's polar inertia and energy while retaining only 36% of its force. Consequently, the solid rod produces 1.64 times more force per unit stored rotational energy. The tube remains advantageous where reduced total mass, lower bearing load, shorter spin-up time under a fixed drive torque, or reduced absolute energy are the controlling constraints.

Central result

At fixed outside diameter, length, material density, angular speed, field orientation, and coupling, hollowing the rotor reduces total K-Field force in direct proportion to the removed mass. It does not change force per unit rotating mass. The solid rod is superior in total force and force per stored spin energy; the tube is superior in low absolute mass, low absolute inertia, and rapid spin-up under a fixed available drive torque.

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1. Scope, assumptions, and coordinate lock

The comparison concerns two complete, coaxial, axisymmetric carbon-fiber rotors. Both have an outside diameter of 10 mm and a length of 100 mm. The tube has a 1 mm radial wall, so its inside diameter is 8 mm. The rod is solid. Both rotate about their longitudinal axes at 100,000 rpm.

A common working density of 1600 kg/m^3 is imposed on both geometries. This density is a declared calculation input. The comparison isolates geometry; it does not introduce different resin fractions, fiber architectures, void fractions, or material coupling coefficients between the two bodies.

The coordinate frame is ICRS/J2000 Cartesian. The Earth CMB traversal direction is $\hat{c}$. The Cell3 tensor G_{phase} transforms the Earth velocity V_E into the effective response vector g .

$$\begin{aligned} g &= G_{\text{phase}} V_E \\ F &= (\kappa_F m \omega / 2) [\hat{n} \text{ cross } g] \\ a &= F/m = (\kappa_F \omega / 2) [\hat{n} \text{ cross } g] \end{aligned}$$

The adopted optimum orientation chooses $\hat{n}$ perpendicular to g and within the plane spanned by $\hat{c}$ and g . The resulting force direction is perpendicular to both $\hat{c}$ and g . This is the maximum pure-lateral orientation.

Quantity	Value	Unit or role
Earth CMB unit vector $\hat{c}_{\text{hat}}$	(-0.970770855836, 0.207348009111, -0.120874929482)	unitless ICRS
Earth velocity V_E	(-359003.199066, 76679.886034, -44701.060101)	m/s ICRS
$G_{\text{phase}} V_E = g$	(-880221.981635, -143911.436562, 266335.026949)	m/s effective response
Magnitude $ g $	930825.217270519235	m/s
κ_F	3.0740508997020345e-11	calibrated coupling
Optimum spin axis $\hat{n}_{\text{opt}}$	(-0.315865036408, 0.646162683024, -0.694768354089)	unitless ICRS
Force direction F_{hat}	(0.077469585421, 0.747375863362, 0.659869519070)	unitless ICRS

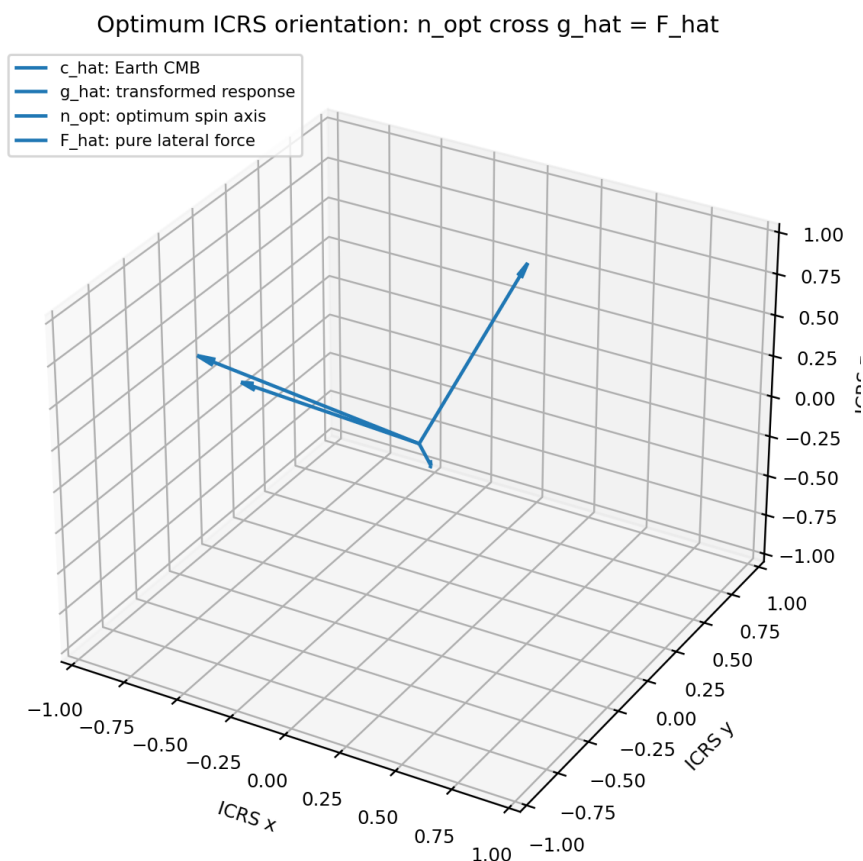


Figure 1. Exact ICRS vector lock. The adopted vectors satisfy $\hat{n}_{\text{opt}} \cdot \hat{g}_{\text{hat}} = 0$, $F_{\text{hat}} \cdot \hat{c}_{\text{hat}} = 0$, $F_{\text{hat}} \cdot \hat{g}_{\text{hat}} = 0$, and $\hat{n}_{\text{opt}} \text{ cross } \hat{g}_{\text{hat}} = F_{\text{hat}}$ to numerical precision.

2. Authoritative Cell3 base cell and face planes

The rotating-body comparison uses the same fixed Cell3 geometry as the broader K-Field force program. The basis matrix B_{Cell3} contains the physical basis vectors a , b , and c as columns. Address coordinates q map to ICRS physical coordinates through $r = B_{\text{Cell3}} q$.

```
a = (0.020381438516, -0.003416414496, -0.002332965916)
b = (0.002198745968, 0.012902170258, -0.034066501536)
c = (-0.000630379486, -0.024257722018, -0.042924354572)
B_Cell3 = [a b c]
```

Authoritative centered Cell3: six faces and all 27 half-step nodes

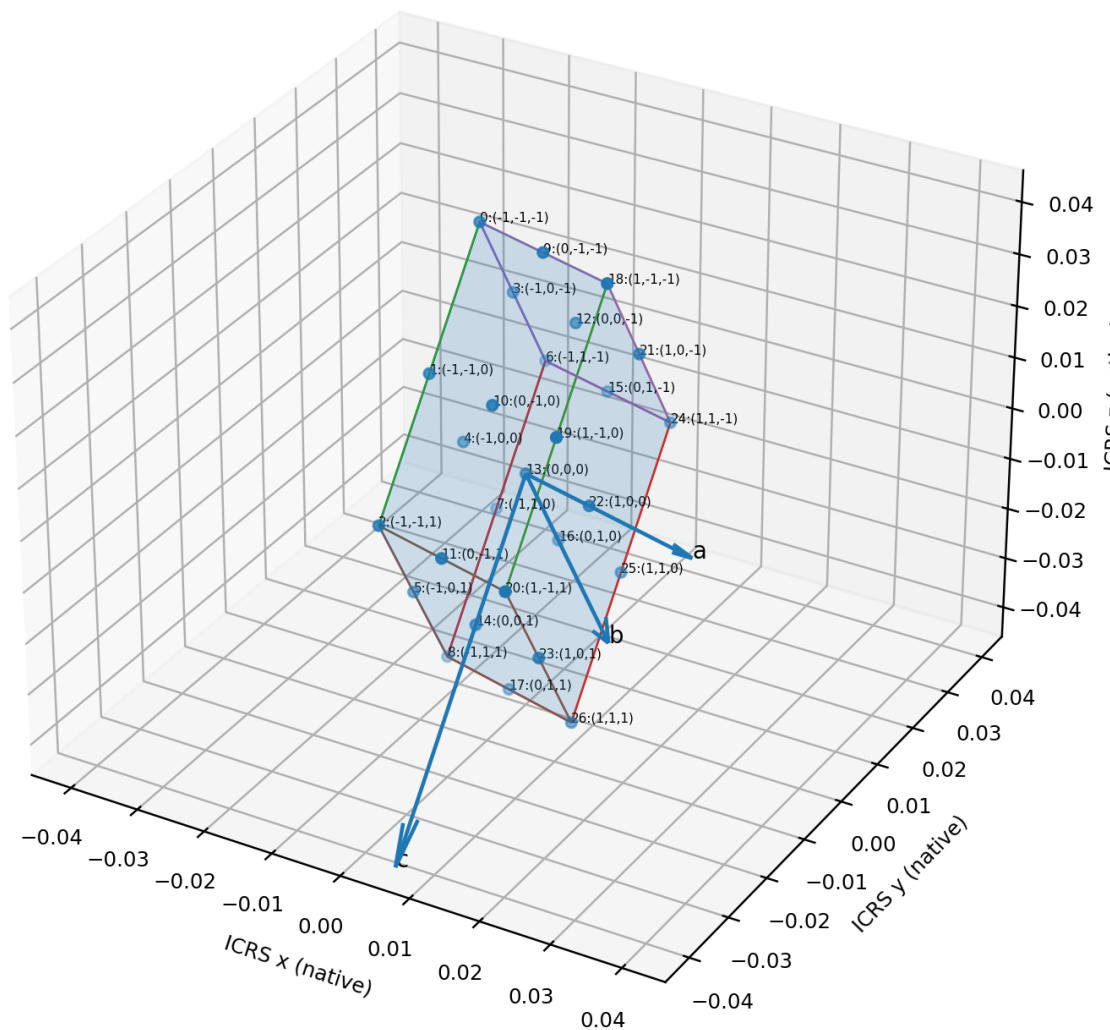


Figure 2. Dimensionally correct centered Cell3. All six boundary faces, 12 edges, 27 centered half-step nodes, integer address labels, and the three basis vectors are generated directly from the numerical ledger.

Cell3 central face-plane families ab, ac, and bc

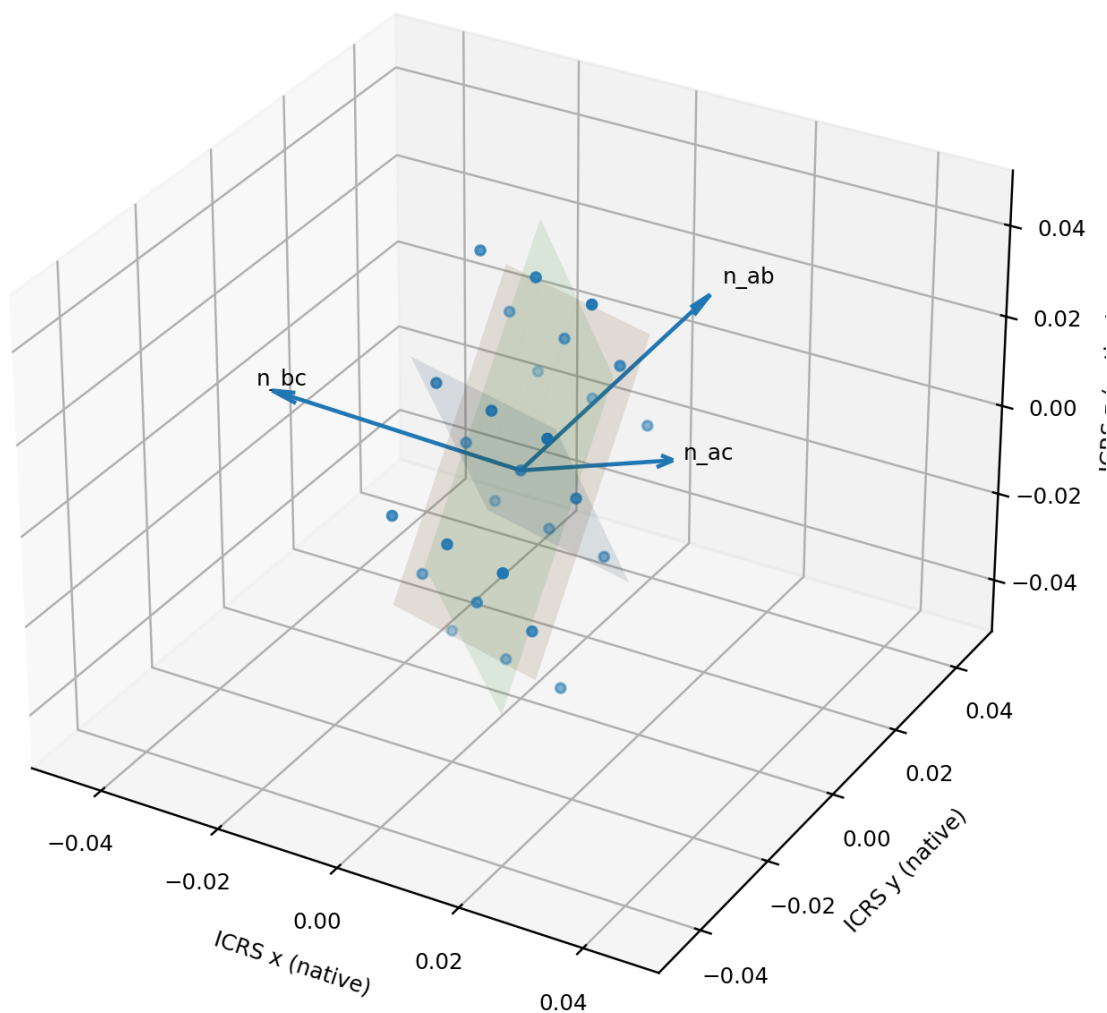


Figure 3. The three central plane families ab, ac, and bc. Each plane is spanned by its corresponding basis pair and carries the listed fixed ICRS normal.

Family	Area (native ²)	Included angle	Unit normal in ICRS
A_ab	7.547217896293e-04	83.933483014 deg	(0.1940922008, 0.9131771688, 0.3583792348)
A_ac	1.011255757668e-03	80.448130021 deg	(0.0890525941, 0.8665767693, -0.4910339483)
A_bc	1.385784431744e-03	50.363231813 deg	(-0.9959651903, 0.0836021625, -0.0326192906)

The base cell is included because the force coefficient and transformed Earth vector are defined in the same fixed ICRS frame. The rotor axis is a device vector in this frame; it is not one of the Cell3 edge vectors and must not be silently substituted for a, b, or c.

3. Rotating-body constitutive law

For a complete circular rotor, integration around the full azimuth removes the rotating point-force pulse and leaves a steady center-of-mass force. Under the strict integrated law used here, the complete rotor force is

$$F = (\kappa_F m \omega / 2) [\hat{n} \text{ cross } (G_{\text{phase}} V_E)].$$

At the optimum orientation, $|\hat{n} \text{ cross } g| = |g|$. Therefore

$$\begin{aligned} |F|_{\text{max}} &= m q_g \omega \\ |F|_{\text{max}}/m &= q_g \omega \\ q_g &= \kappa_F |g| / 2 \end{aligned}$$

The calculated coefficient is $q_g = 1.430702048307891\text{e-}05$ m/s per radian of rotation. At 100,000 rpm, $\omega = 10471.975511965977$ rad/s and the maximum specific force is 0.149822768148 N/kg.

The law is linear in total rotating mass and angular speed. Polar moment of inertia does not appear in the force expression as an independent multiplier. Inertia enters the mechanical cost of producing and maintaining the angular speed.

Reversing the sign of ω reverses the complete force vector by 180 deg. Reversing the physical axis polarity while preserving the same directed angular velocity produces the same physical spin state; sign conventions must therefore be bound to the right-hand rule.

4. Common rotor dimensions and material ledger

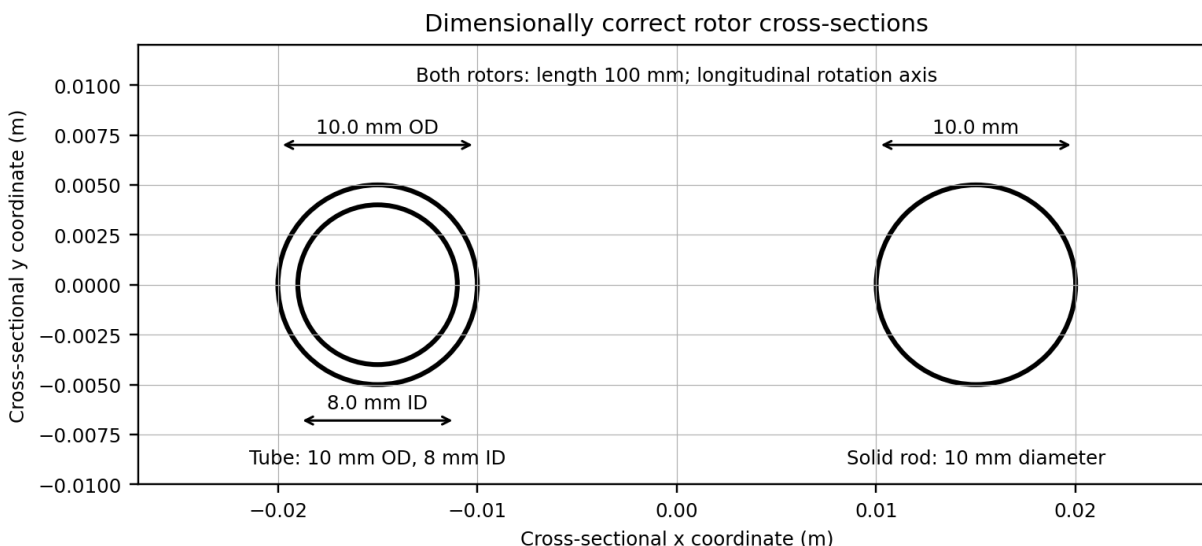


Figure 4. Exact cross-sections. Both bodies have 5 mm outer radius and 100 mm length. The tube has a 4 mm bore radius and 1 mm radial wall.

Input	Tube	Solid rod	Unit
Length L	0.100000	0.100000	m
Outer diameter D_o	0.010000	0.010000	m
Outer radius R_o	0.005000	0.005000	m
Inner radius R_i	0.004000	0.000000	m
Wall thickness	0.001000	not applicable	m
Working density rho	1600.000	1600.000	kg/m ³
Spin rate	100000.000	100000.000	rpm
Spin frequency	1666.666666667	1666.666666667	Hz
Angular speed omega	10471.975511965977	10471.975511965977	rad/s
Outer rim speed	52.359877559830	52.359877559830	m/s

5. Tube mass, inertia, and force derivation

The tube material occupies the annular cross-sectional area

$$A_{\text{tube}} = \pi (R_o^2 - R_i^2).$$

Its volume and mass are

$$\begin{aligned} V_{\text{tube}} &= \pi (R_o^2 - R_i^2) L \\ m_{\text{tube}} &= \rho \pi (R_o^2 - R_i^2) L \end{aligned}$$

Substitution gives $V_{\text{tube}} = 2.827433388230815 \times 10^{-6} \text{ m}^3$ and $m_{\text{tube}} = 0.004523893421169 \text{ kg} = 4.523893421 \text{ g}$.

At optimum orientation, the force is

$$F_{\text{tube}} = m_{\text{tube}} q_g \omega.$$

The result is $F_{\text{tube}} = 6.777822351661018 \times 10^{-4} \text{ N} = 0.677782235 \text{ mN}$. The equivalent static gram-force is 0.069114553 gf .

The axial polar moment of a uniform annular cylinder is

$$I_{\text{tube}} = (1/2) m_{\text{tube}} (R_o^2 + R_i^2).$$

$I_{\text{tube}} = 9.273981513397073 \times 10^{-8} \text{ kg m}^2$. At the stated angular speed, $H_{\text{tube}} = I_{\text{tube}} \omega = 9.711690730671932 \times 10^{-4} \text{ kg m}^2/\text{s}$ and $E_{\text{tube}} = (1/2) I_{\text{tube}} \omega^2 = 5.085029375569 \text{ J}$.

6. Solid-rod mass, inertia, and force derivation

The solid rod occupies the complete disk area

$$A_{\text{rod}} = \pi R_o^2.$$

Its volume and mass are

$$\begin{aligned} V_{\text{rod}} &= \pi R_o^2 L \\ m_{\text{rod}} &= \rho \pi R_o^2 L \end{aligned}$$

Substitution gives $V_{\text{rod}} = 7.853981633974484 \times 10^{-6} \text{ m}^3$ and $m_{\text{rod}} = 0.012566370614359 \text{ kg} = 12.566370614 \text{ g}$.

The optimum force is

$$F_{\text{rod}} = m_{\text{rod}} q_g \omega.$$

The result is $F_{\text{rod}} = 1.882728431016949 \times 10^{-3} \text{ N} = 1.882728431 \text{ mN}$. The equivalent static gram-force is 0.191984871 gf .

The axial polar moment of a uniform solid cylinder is

$$I_{\text{rod}} = (1/2) m_{\text{rod}} R_o^2.$$

$I_{\text{rod}} = 1.570796326794897 \times 10^{-7} \text{ kg m}^2$. The angular momentum is $H_{\text{rod}} = 1.644934066848226 \times 10^{-3} \text{ kg m}^2/\text{s}$ and the stored spin energy is $E_{\text{rod}} = 8.612854633417 \text{ J}$.

7. Direct numerical comparison at 100,000 rpm

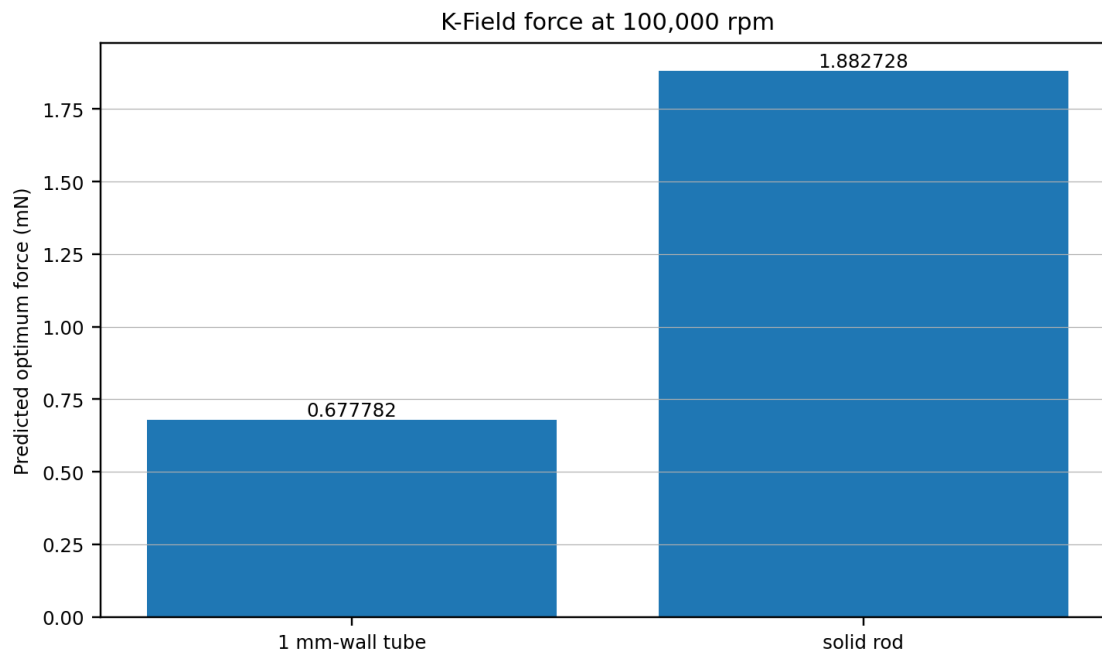


Figure 5. The solid rod generates 2.777778 times the total force because it contains 2.777778 times the rotating mass.

Quantity	1 mm-wall tube	Solid rod	Solid / tube
Rotating mass	4.523893421 g	12.566370614 g	2.777777778
Predicted force	0.677782235 mN	1.882728431 mN	2.777777778
Apparent-weight equivalent	0.069114553 gf	0.191984871 gf	2.777777778
Force per rotating mass	0.149822768148 N/kg	0.149822768148 N/kg	1.000000000
Polar moment	9.273981513397e-08	1.570796326795e-07	1.693766938
Angular momentum	9.711690730672e-04	1.644934066848e-03	1.693766938
Stored spin energy	5.085029376 J	8.612854633 J	1.693766938
Force per stored energy	1.332897383882e-04 N/J	2.185951709567e-04 N/J	1.640000000

The force ratio is exactly the area and mass ratio for equal density and length:

$$\begin{aligned}
 F_{\text{rod}}/F_{\text{tube}} &= m_{\text{rod}}/m_{\text{tube}} \\
 &= R_o^2 / (R_o^2 - R_i^2) \\
 &= 25 / 9 \\
 &= 2.777777778.
 \end{aligned}$$

8. Why force per unit rotating mass is identical

The complete integrated force law contains total mass as a linear prefactor. Dividing either rotor's force by its own rotating mass removes geometry and density:

$$\begin{aligned} F_{\text{tube}}/m_{\text{tube}} &= q_g \omega \\ F_{\text{rod}}/m_{\text{rod}} &= q_g \omega. \end{aligned}$$

Both values equal 0.149822768148 N/kg. Hollowing the body removes mass and force in the same proportion. It does not improve or degrade force per unit rotating mass at fixed angular speed.

This conclusion differs from the small-radius comparison at fixed rim speed. In that different comparison, $\omega = v_{\text{rim}}/R$ and reducing the entire rotor radius increases angular speed. Here the outside radius and rpm are identical. The tube and rod therefore have the same ω , and no $1/R$ enhancement distinguishes them.

The material inside a solid rod has different local tangential speeds, but every concentric cylindrical shell shares the same angular rate. Under the strict current law, each shell has the same force per unit shell mass because the shell response is controlled by ω rather than by its local radius as an independent factor.

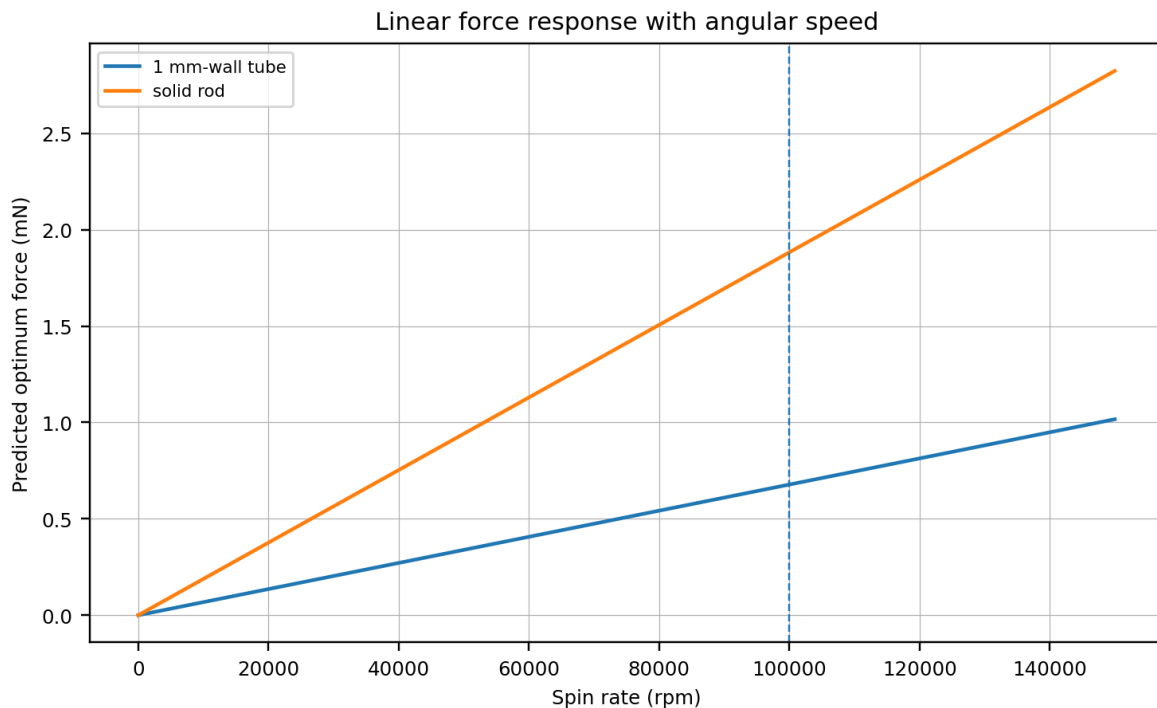


Figure 6. Both force curves are linear in rpm. Their slope ratio is constant and equal to the mass ratio.

9. Concentric-shell derivation of the solid rod

A solid rod can be decomposed into thin coaxial cylindrical shells of radius r and thickness dr . The mass of one shell is

$$dm = \rho (2 \pi r L dr).$$

Applying the same integrated circular law to the shell gives

$$dF = q_g \omega dm.$$

Integration from $r = 0$ to R_o gives

$$\begin{aligned} F_{\text{rod}} &= \int_0^{R_o} q_g \omega \rho 2 \pi r L dr \\ &= q_g \omega \rho \pi R_o^2 L \\ &= q_g \omega m_{\text{rod}}. \end{aligned}$$

The force contribution inside radius r is proportional to r^2 . The polar-inertia contribution inside radius r is proportional to r^4 . Therefore the outer annulus carries a larger fraction of spin energy than it carries of mass or K-Field force.

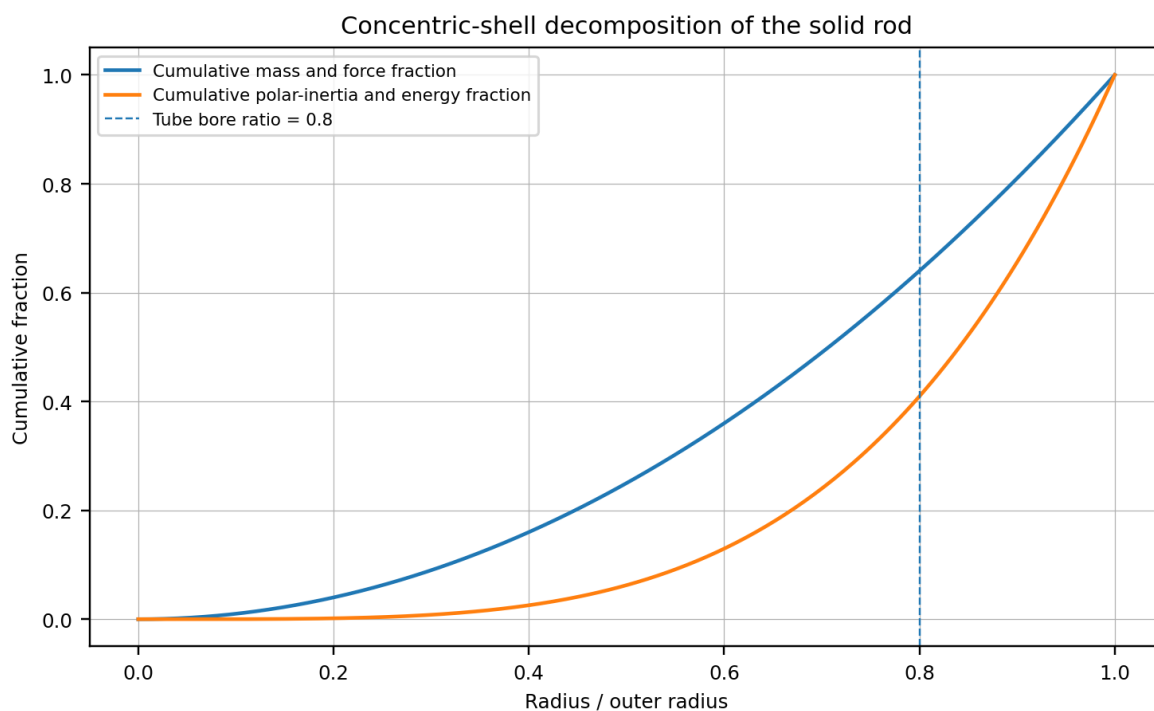


Figure 7. At the tube bore ratio $R_i/R_o = 0.8$, the removed central core contains 64% of the solid rod's mass and force but only 40.96% of its polar inertia and stored spin energy. The remaining annulus contains 36% of force and 59.04% of energy.

10. Stored energy and angular momentum

The force law does not contain polar moment as an independent factor, but the mechanical state required to establish the rotation does. At a common angular speed,

$$H = I \omega$$

$$E = (1/2) I \omega^2.$$

The tube-to-solid inertia ratio is

$$I_{\text{tube}}/I_{\text{rod}}$$

$$= [(R_o^2 - R_i^2)(R_o^2 + R_i^2)] / R_o^4$$

$$= 1 - (R_i/R_o)^4$$

$$= 0.5904.$$

The same ratio applies to angular momentum and spin energy at the common angular speed. The tube requires 59.04% of the solid rod's angular momentum and stored energy, not 36%.

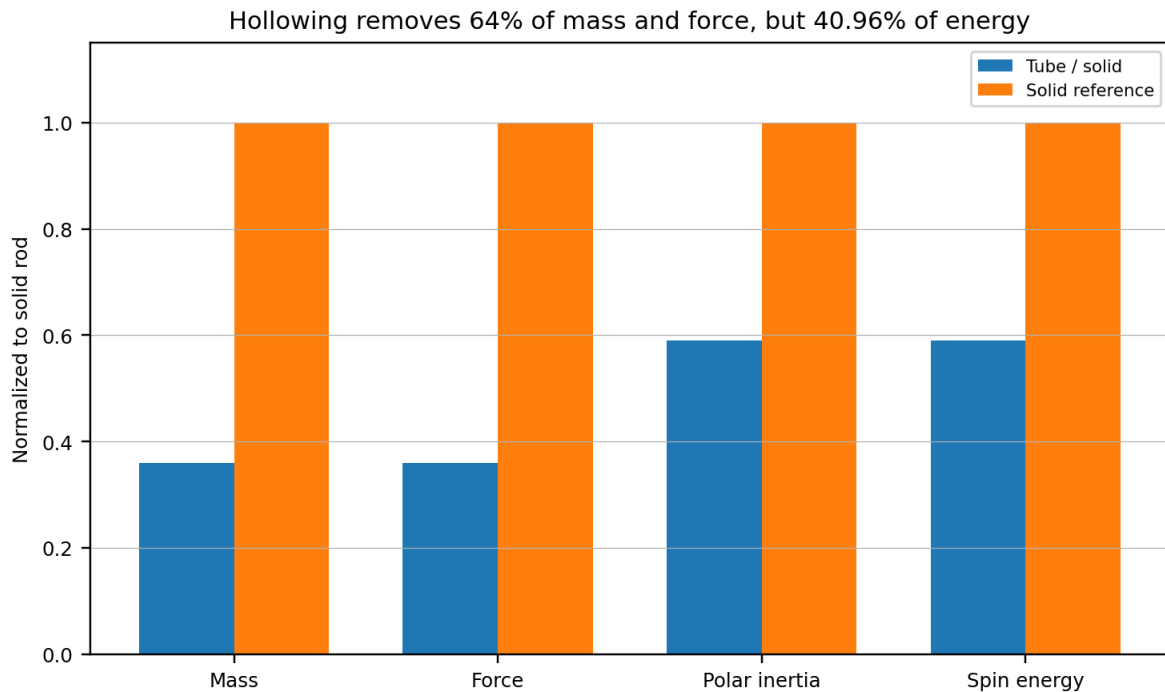


Figure 8. Force follows mass, while spin energy follows polar inertia. Hollowing preferentially removes low-radius material that contributes little to inertia but contributes normally to mass and force.

11. Force per stored rotational energy

For the tube,

$$F_{\text{tube}}/E_{\text{tube}}$$

$$= [m_{\text{tube}} q_g \omega] / [(1/4)m_{\text{tube}}(R_o^2 + R_i^2)\omega^2]$$

$$= 4 q_g / [(R_o^2 + R_i^2) \omega].$$

For the solid rod,

$$F_{\text{rod}}/E_{\text{rod}}$$

$$= [m_{\text{rod}} q_g \omega] / [(1/4)m_{\text{rod}} R_o^2 \omega^2]$$

$$= 4 q_g / [R_o^2 \omega].$$

The solid-to-tube advantage is therefore

$$(F/E)_{\text{rod}} / (F/E)_{\text{tube}}$$

$$= (R_o^2 + R_i^2) / R_o^2$$

$$= 1 + (R_i/R_o)^2$$

$$= 1.64.$$

For this exact geometry and fixed rpm, the solid rod produces 1.64 times more K-Field force per joule of stored rotational energy. The reason is geometric: central material contributes fully to force but carries little rotational energy because its local radius is small.

12. Hollowing laws at fixed outside diameter and rpm

Let $x = R_i/R_o$ be the normalized bore radius. Relative to a solid rod with the same outer radius, length, density, and angular speed,

Mass fraction	$= 1 - x^2$
Force fraction	$= 1 - x^2$
Polar-inertia fraction	$= 1 - x^4$
Spin-energy fraction	$= 1 - x^4$
Specific-force fraction	$= 1$
Force/energy fraction	$= 1 / (1 + x^2)$

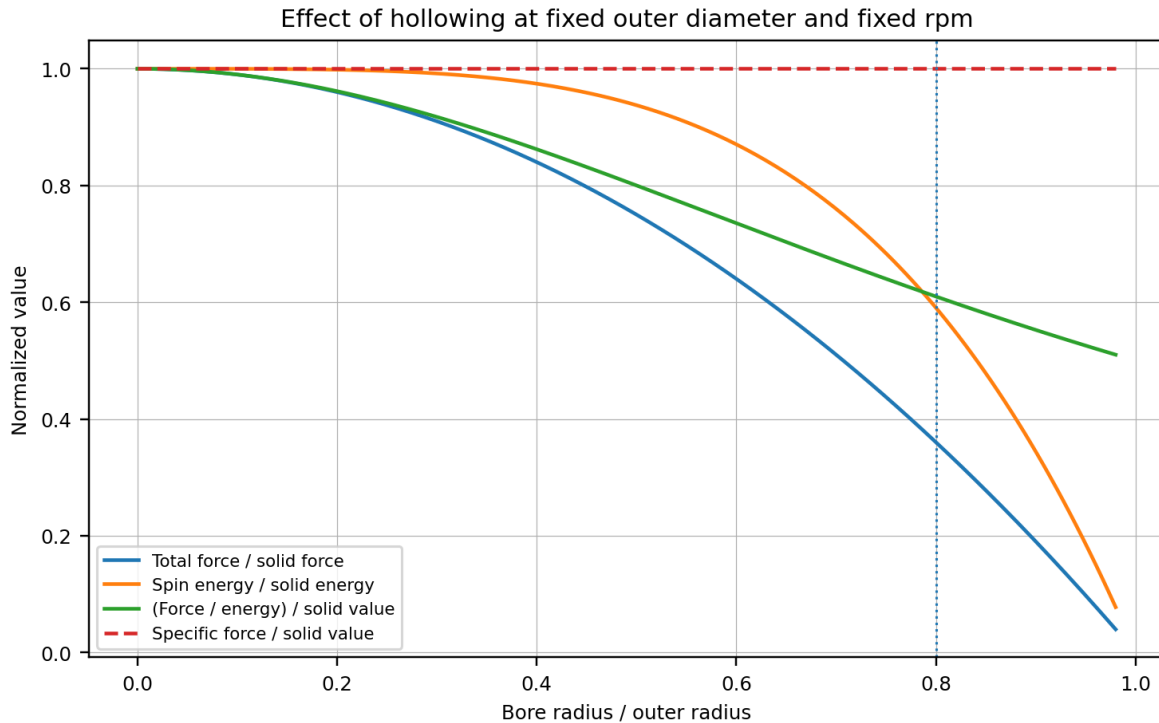


Figure 9. Hollowing at fixed outer diameter and rpm. Specific force remains exactly constant, but total force declines faster than stored spin energy. The vertical line marks the present $x = 0.8$ tube.

A thin-walled tube is therefore not automatically the most efficient geometry under every definition. It is efficient in mass reduction and bearing-load reduction. It is not efficient in total force per fixed outside envelope or in force per stored rotational energy at fixed rpm.

13. Drive torque, spin-up time, and steady rotation

Under a constant applied spin-up torque τ_d and negligible losses, angular acceleration is $\alpha = \tau_d/I$. The time to reach the target angular speed is

$$t_{\text{spin}} = I \omega / \tau_d.$$

Because $I_{\text{tube}}/I_{\text{rod}} = 0.590400$, the tube reaches 100,000 rpm in 59.04% of the solid rod's spin-up time under the same constant drive torque.

Conversely, under the same allowed spin-up time, the tube requires 59.04% of the solid rod's ideal acceleration torque. These are meaningful system advantages where motor torque, magnetic-bearing authority, or startup time is limiting.

Once the target rpm is reached, ideal rigid-body mechanics requires no power merely to maintain rotation. Actual steady drive power is set by aerodynamic drag, bearing losses, internal material damping, electromagnetic losses, and control-system work. Those losses are apparatus-dependent and are not inferred from polar inertia alone.

The solid rod's larger force does not require 2.777778 times the stored energy; it requires 1.693767 times the stored energy. This is why the solid rod is better on force per stored-energy basis even though the tube is easier to accelerate in absolute terms.

14. Constant-rim-speed radius scaling

The tube-versus-rod comparison above holds the same outer radius and the same rpm. A separate scaling question holds rim speed constant while changing the complete rotor radius. Then

$$\begin{aligned} \omega &= v_{\text{rim}}/R \\ F/m &= q_g v_{\text{rim}}/R. \end{aligned}$$

Under that constraint, smaller radius increases force per unit rotating mass because it increases the cycle rate. This is the origin of the $1/R$ small-radius advantage. It does not distinguish a hollow tube from a solid rod of the same outer radius at the same rpm.

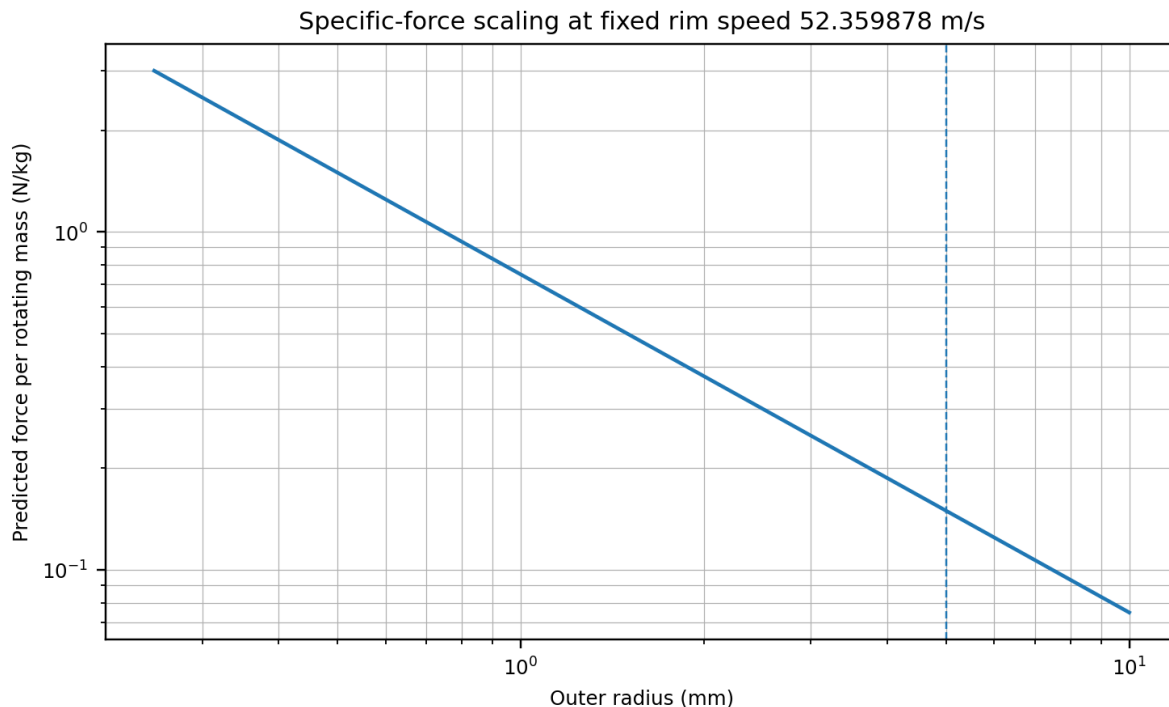


Figure 10. Specific-force scaling at the present outer-rim speed of 52.359878 m/s. The 5 mm radius point reproduces 0.149822768148 N/kg. Reducing the complete radius while preserving rim speed raises ω and the specific force.

The two geometric optimizations can be combined only by changing the outside radius: use the smallest radius compatible with material coherence, balance, drive bandwidth, and loss control, then decide whether the cross-section should be solid or hollow according to total force, energy, mass, and bearing constraints.

15. Experimental observables and falsifiable distinctions

A direct comparison should mount the tube and solid rod in the same bearing, drive, vacuum, thermal, and force-balance architecture. The spin axis should be set to n_{opt} . The two bodies should be separately balanced to the same residual mechanical vibration standard.

The strict model predicts the following simultaneous signatures:

Observable	Predicted tube/solid relation
Force magnitude	$F_{tube}/F_{rod} = 0.36$
Force direction	Identical for equal spin sign and axis orientation
Spin reversal	Both force vectors reverse by 180 deg
Specific force	Equal: $F_{tube}/m_{tube} = F_{rod}/m_{rod}$
Force versus rpm	Linear for each; slope ratio = 0.36
Stored spin energy	$E_{tube}/E_{rod} = 0.5904$
Spin-up time at equal constant drive torque	$t_{tube}/t_{rod} = 0.5904$
Strict geometric null	Both vanish when n_{hat} is parallel to g

The strongest comparison is not merely the absolute force ratio. It is the joint observation that total force follows the 0.36 mass fraction while specific force remains equal and stored energy follows the different 0.5904 inertia fraction.

Departures discriminate mechanism classes. If force follows polar inertia rather than mass, the tube-to-solid force ratio would approach 0.5904. If force is dominated by outer surface area, the two bodies could approach equal force despite different mass. If only a surface layer is active, force will depend on wall thickness and material penetration depth. The present mass-linear law selects the 0.36 ratio.

16. Conclusions

At 100,000 rpm and optimum orientation, the 10 mm outside-diameter x 100 mm carbon-fiber tube with a 1 mm wall generates 0.677782 mN using the declared density of 1600 kg/m^3 . The solid rod of the same outside dimensions generates 1.882728 mN.

The solid rod produces 2.777778 times more total force because it contains 2.777778 times more rotating mass. Both bodies produce exactly the same force per unit rotating mass, 0.149822768 N/kg, under the strict mass-linear force law.

Hollowing does not improve specific force at fixed outer radius and rpm. It reduces total force and total mass by the same factor. It does reduce absolute inertia, energy, angular momentum, drive torque, and spin-up time.

The solid rod is superior in force per stored rotational energy by a factor of 1.64 because low-radius core material contributes normally to force while contributing weakly to polar inertia. The tube is preferable when absolute mass, bearing load, startup torque, or spin-up time dominates system design.

The independent small-radius advantage remains valid when the entire outside radius is reduced at constant rim speed. That optimization is controlled by $\omega = v_{rim}/R$ and is distinct from hollowing a fixed-diameter rotor.

For the exact 10 mm geometry studied here, the solid rod is the force-maximizing and force-per-energy-maximizing cross-section. The 1 mm-wall tube is the lower-mass, lower-inertia, faster-spin-up cross-section.

Appendix A. Complete numerical input and result ledger

Field	Tube	Solid rod	Unit
Working density	1600	1600	kg/m ³
Length	0.1	0.1	m
Outer radius	0.005	0.005	m
Inner radius	0.004	0	m
Cross-sectional area	2.827433388230814e-05	7.853981633974483e-05	m ²
Volume	2.827433388230815e-06	7.853981633974484e-06	m ³
Mass	4.523893421169304e-03	1.256637061435917e-02	kg
Spin rate	100000	100000	rpm
Angular speed	1.047197551196598e+04	1.047197551196598e+04	rad/s
Outer rim speed	5.235987755982988e+01	5.235987755982988e+01	m/s
Specific force	1.498227681479979e-01	1.498227681479979e-01	N/kg
Total force	6.777822351661018e-04	1.882728431016949e-03	N
Polar moment	9.273981513397073e-08	1.570796326794897e-07	kg m ²
Angular momentum	9.711690730671932e-04	1.644934066848226e-03	kg m ² /s
Spin energy	5.085029375569172e+00	8.612854633416616e+00	J
Force per energy	1.332897383882344e-04	2.185951709567044e-04	N/J

Field vector ledger

Object	x	y	z	Magnitude
c_hat	-0.970770855836	0.207348009111	-0.120874929482	1.000000000000
V_E (m/s)	-359003.199066	76679.886034	-44701.060101	369812.502001
g (m/s)	-880221.981635	-143911.436562	266335.026949	930825.217271
g_hat	-0.945636157362	-0.154606293310	0.286127859460	1.000000000000
n_opt	-0.315865036408	0.646162683024	-0.694768354089	1.000000000000
F_hat	0.077469585421	0.747375863362	0.659869519070	1.000000000000

Appendix B. Complete 27-node centered Cell3 registry

#	2i	2j	2k	lambda	ICRS x	ICRS y	ICRS z
0	-1	-1	-1	-3	-0.010974902499	0.007385983128	0.039661911012
1	-1	-1	0	-2	-0.011290092242	-0.004742877881	0.018199733726
2	-1	-1	1	-1	-0.011605281985	-0.016871738890	-0.003262443560
3	-1	0	-1	-2	-0.009875529515	0.013837068257	0.022628660244
4	-1	0	0	-1	-0.010190719258	0.001708207248	0.001166482958
5	-1	0	1	0	-0.010505909001	-0.010420653761	-0.020295694328
6	-1	1	-1	-1	-0.008776156531	0.020288153386	0.005595409476
7	-1	1	0	0	-0.009091346274	0.008159292377	-0.015866767810
8	-1	1	1	1	-0.009406536017	-0.003969568632	-0.037328945096
9	0	-1	-1	-2	-0.000784183241	0.005677775880	0.038495428054
10	0	-1	0	-1	-0.001099372984	-0.006451085129	0.017033250768
11	0	-1	1	0	-0.001414562727	-0.018579946138	-0.004428926518
12	0	0	-1	-1	0.000315189743	0.012128861009	0.021462177286
13	0	0	0	0	0.000000000000	0.000000000000	0.000000000000
14	0	0	1	1	-0.000315189743	-0.012128861009	-0.021462177286
15	0	1	-1	0	0.001414562727	0.018579946138	0.004428926518
16	0	1	0	1	0.001099372984	0.006451085129	-0.017033250768
17	0	1	1	2	0.000784183241	-0.005677775880	-0.038495428054
18	1	-1	-1	-1	0.009406536017	0.003969568632	0.037328945096
19	1	-1	0	0	0.009091346274	-0.008159292377	0.015866767810
20	1	-1	1	1	0.008776156531	-0.020288153386	-0.005595409476
21	1	0	-1	0	0.010505909001	0.010420653761	0.020295694328
22	1	0	0	1	0.010190719258	-0.001708207248	-0.001166482958
23	1	0	1	2	0.009875529515	-0.013837068257	-0.022628660244
24	1	1	-1	1	0.011605281985	0.016871738890	0.003262443560
25	1	1	0	2	0.011290092242	0.004742877881	-0.018199733726
26	1	1	1	3	0.010974902499	-0.007385983128	-0.039661911012

Appendix C. Cell3 boundary vertices and face families

#	i	j	k	ICRS x	ICRS y	ICRS z
0	-0.5	-0.5	-0.5	-0.010974902499	0.007385983128	0.039661911012
1	-0.5	-0.5	0.5	-0.011605281985	-0.016871738890	-0.003262443560
2	-0.5	0.5	-0.5	-0.008776156531	0.020288153386	0.005595409476
3	-0.5	0.5	0.5	-0.009406536017	-0.003969568632	-0.037328945096
4	0.5	-0.5	-0.5	0.009406536017	0.003969568632	0.037328945096
5	0.5	-0.5	0.5	0.008776156531	-0.020288153386	-0.005595409476
6	0.5	0.5	-0.5	0.011605281985	0.016871738890	0.003262443560
7	0.5	0.5	0.5	0.010974902499	-0.007385983128	-0.039661911012

Family	Area (native^2)	Included angle	Unit normal in ICRS
A_ab	7.547217896293e-04	83.933483014 deg	(0.1940922008, 0.9131771688, 0.3583792348)
A_ac	1.011255757668e-03	80.448130021 deg	(0.0890525941, 0.8665767693, -0.4910339483)
A_bc	1.385784431744e-03	50.363231813 deg	(-0.9959651903, 0.0836021625, -0.0326192906)

Basis-vector details

Vector	ICRS x	ICRS y	ICRS z	Length
a	0.020381438516	-0.003416414496	-0.002332965916	0.020797058781
b	0.002198745968	0.012902170258	-0.034066501536	0.036494205130
c	-0.000630379486	-0.024257722018	-0.042924354572	0.049308565901

Appendix D. Compact equation set and reconstruction procedure

Relationship	Equation
Tube mass	$m_{\text{tube}} = \rho \pi (R_o^2 - R_i^2) L$
Solid-rod mass	$m_{\text{rod}} = \rho \pi R_o^2 L$
Angular speed	$\omega = (2 \pi / 60) \text{ RPM}$
Field response	$g = G_{\text{phase}} V_E$
Rotor force	$F = (\kappa_F m \omega / 2) \hat{n} \times g$
Free acceleration	$a = F/m = (\kappa_F \omega / 2) \hat{n} \times g$
Optimum axis	$\hat{n}_{\text{opt}} = \text{normalized component of } \hat{c}_{\text{hat}} \text{ perpendicular to } g$
Force direction	$\hat{F}_{\text{hat}} = (\hat{c}_{\text{hat}} \times g) / \hat{c}_{\text{hat}} \times g $
Maximum specific force	$q_g \omega$, where $q_g = \kappa_F g / 2$
Tube polar moment	$I_{\text{tube}} = (1/2) m_{\text{tube}} (R_o^2 + R_i^2)$
Solid polar moment	$I_{\text{rod}} = (1/2) m_{\text{rod}} R_o^2$
Angular momentum	$H = I \omega$
Spin energy	$E = (1/2) I \omega^2$
Force ratio	$F_{\text{rod}}/F_{\text{tube}} = R_o^2 / (R_o^2 - R_i^2)$
Specific-force ratio	$(F/m)_{\text{rod}} / (F/m)_{\text{tube}} = 1$
Force/energy advantage	$(F/E)_{\text{rod}} / (F/E)_{\text{tube}} = (R_o^2 + R_i^2)/R_o^2$

Minimal reconstruction sequence

1. Load G_{phase} , V_E , $\hat{c}_{\text{hat}}$, and κ_F .
2. Compute $g = G_{\text{phase}} V_E$, $\hat{g}_{\text{hat}} = g/|g|$, $\hat{n}_{\text{opt}}$, and $\hat{F}_{\text{hat}}$.
3. Convert 100,000 rpm to $\omega = 10471.975511966 \text{ rad/s}$.
4. Compute tube and rod volumes and masses from exact circular geometry.
5. Compute the common specific force $q_g \omega$.
6. Multiply the common specific force by each rotating mass.
7. Compute each polar moment, angular momentum, and stored spin energy.
8. Form ratios independently from geometry and verify they reproduce the direct numerical calculations.
9. Reverse ω to reverse the complete force vector without changing magnitude.

Appendix E. Archival record

Field	Record
Document title	K-Field Solid Rod Versus Tube
Principal Investigator	P. Dwayne Esterline
Education	BS, Huntington University (1999); MBA, Indiana Wesleyan University (2008)
Organization	Krytronx
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Generation UTC	2026-07-26T18:42:23Z
Generation epoch	1785091343
Output PDF	K-Field_solid_rod_vs_tube_1785091343.pdf
Machine-readable companion	K-Field_solid_rod_vs_tube_1785091343.json
Report generation	Python ReportLab; built-in core fonts only
Figure generation	Python numerical computation and raster rendering from the exact report ledger
Coordinate frame	ICRS/J2000 Cartesian
Mechanical units	SI
Cell3 units	native source units
Source lineage	K-Field tubular rotor free-space force report; Cell3 starting-orientation report; linear-inertia derivation

P. Dwayne Esterline

Krytronx

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